

Evolving Turmites with Gaussian Genomes

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Introduction

There is a popular meme on the Internet that stems from the concept of “carcinization,” a term for the convergent evolution of various decapods taking on crab-like morphology. The term was first introduced by the English zoologist Lancelot Alexander Borradaile [7] in his paper, “*Porcellanopagurus*: an instance of carcinization” where he uses the term to describe the crabbiness of a genus of hermit crab [2]. There have been various subsequent papers adding onto the catalog of decapods who have evolved to look and act like crabs, such as Keiler et al.’s “One hundred years of carcinization – the evolution of the crab-like habitus in *Anomura*” [4] that points to multiple independent events of carcinization. People of the Internet looked at these papers and jokingly asserted that crabs are the ultimate life form and all life will converge to becoming crabs.

The meme itself is a total misconception on how evolution works in practice. This is because evolution tends to favor variance over convergence due to competition over food and other niches, but to me it poses an interesting question about the Central Limit Theorem (CLT). Recall the way we defined the CLT in class:

For i.i.d. random variables $X_1, X_2, \dots$ with finite mean μ and finite variance σ^2 ,

$$\frac{\sqrt{n}}{\sigma} \left(\frac{X_1 + \dots + X_n}{n} - \mu \right) \xrightarrow{d} \mathcal{N}(0, 1).$$

If we think about evolution and life in a simplified and albeit extremely reductionist sense, the actions and decisions an organism makes to express itself in the world are samples from a “genome” determined by a random distribution. Sexual reproduction is essentially a process of creating that random distribution via the averaging of the parent’s individual distributions plus a little bit of noise from mutations. So, without the pressures of finite resources, can we create a model where evolution does

converge? We will get into the specifics about the model itself later, but from this basic description, we have a supposed infinitely large average of a bunch of random variables, the only problem is that these random variables are not i.i.d. by design. Therefore, a natural question to ask is if my naive evolution model is destined to converge to a single distribution like for the CLT? If so, what does changing the parameters of evolution do to convergence?

Relevant Works

I was inspired to take on this project because of a video posted by a user (whose real name I couldn't find) with the username Emergent Garden on YouTube [3]. In the video he showcases a website (<https://evolvecode.io/turmites/index.html>) he “vibecoded” (programmed using code exclusively written by generative AI) to visualize Turmites working in real time. A Turmite is a kind of 2-dimensional Turing machine where the head of the machine acts on a 2-dimensional grid instead of a 1-dimensional tape and can not only move forward and backward, but also can move side to side [5].

Unfortunately, the only work that I could find that uses Turmites for evolutionary systems is behind a paywall that I couldn't resolve in time for submission of this project. It is a paper called “Vants and Turmites” written by Greg Turk whose abstract says they investigate how Turmites can be used to simulate biological phenomena like self-reproduction [6]. Hopefully we will all get access to this paper via Cornell later, but for now I apologize that I didn't manage my time well to have this paper accessible before submission.

Another loosely related paper is a review by Okezue Bell titled “Applications of Gaussian Mutation for Self Adaptation in Evolutionary Genetic Algorithms” that is seemingly incomplete since it has a conclusion header with no text underneath [1]. The main takeaway of this paper is the concept of Gaussian mutations which take real valued variables $x \in [a, b]$ and mutate them with the function

$$M_g(x) := \min(\max(\mathcal{N}(x, \sigma), a), b)$$

This function is very close to the Gaussian mutation perturbations that my model uses. The results of this paper show that via simulations Gaussian mutation can outperform other fixed-parameter approaches to various optimization problems the author investigates like robotic arm trajectory optimization, molecular geometry optimization, and sentiment lexicon classification. Although my experiments aren't particularly concerned with optimization of any kind of task, the prevalence of Gaussian mutation in evolutionary computation literature suggests that encoding a Turmite's genome as a Gaussian random variable is a natural and well-motivated choice.

The Model

My experiments take place on a modified version of Emergent Garden’s website. On the website, Emergent Garden grants the user the ability to write custom sets of rules for the Turmites via JSON formatted files. The user can choose from relative moves based on where the Turmite is currently facing (turn left, turn right, move forward, do a U-turn), absolute moves (move left, right, up, or down on the screen), or they can choose to have the Turmite make a random move drawn as a uniform sample from the moves listed above. Like a traditional Turing machine, the Turmite makes these moves based on the current symbol it is looking at. In this case, Emergent Garden represents the Turmite’s alphabet as 12 different colors. For the sake of simplicity in this experiment, I limited the possible moves to the set of relative moves.

For the evolution model, I give each Turmite an inherent “genome” represented by a Gaussian random variable, from which we repeatedly sample to generate the deterministic ruleset for that Turmite. Then, when two Turmites collide, one of the parents, chosen with a uniform coin toss, will die and be replaced by their child. For a child of two Turmites with genomes $\mathcal{N}(\mu_1, 0.5)$ and $\mathcal{N}(\mu_2, 0.5)$, its genome will be

$$\mathcal{N}\left(\frac{\mu_1 + \mu_2}{2} + \alpha\epsilon, 0.5\right).$$

In this formula, α and ϵ represent the independent variables of the experiment. α is what we will call the “mutation strength”, determining how much we decide to have a mutation affect the creation of our child Turmite. Higher values of α indicate stronger effects from mutations. For ϵ , I don’t actually directly manipulate this variable, instead ϵ is a sample from our “mutation distribution” which is itself a normal distribution $\mathcal{N}(0, \sigma)$ where σ is what we will call the “mutation noise” parameter, which we will directly manipulate. Higher values of σ dictate mutations with higher variance or noise.

When a Turmite is born, its ruleset is first determined by drawing uniform random numbers within the intervals for the max states and colors allowed for a Turmite. In this experiment, I use 1000 and 12 respectively. Then for each state, we sample a uniform random move from the pool of possible moves to perform and we sample a random color to be the color written after performing the move. The color is drawn from the Turmite’s genome Gaussian, which is continuous. This value is made discrete by taking the floor of the sample to give us the integer c , then the color index is chosen via the formula

$$c' = (c \bmod 12 + 12) \bmod 12$$

which ensures that c' is an integer in the interval $[0, 11]$ so it works as a valid index for the array used to generate colors.

Note that in my model of evolution, everything is either drawn from uniform distributions or Gaussian distributions. The driving mechanism for evolution is exclusively Gaussian, and this was highly motivated by the nice properties Gaussians have which were presented in class. Specifically, the fact that linear combinations of Gaussians are themselves Gaussian facilitates an environment where we can cleanly reason about the descendants of the initial Turmites many many iterations after initialization. In the model, a variation of the CLT isn't even necessary because all child genomes will be Gaussian. The question now concerns whether all children will converge to having similar genomes and if so how fast or will divergent genomes emerge forming new species?

Results

I initially conducted 5 experiments to see the effects that each parameter has on convergence rate since I assumed that convergence was a guarantee. To collect data for these experiments, I recorded every time a new Turmite was born along with its exact ruleset and its genome. Data collection was paused after 10000 ticks of the simulation going at a rate of 1000 steps/second. The simulations involved 4 Turmites, all spawning in a square at the center facing uniform random directions to encourage mating. The 5 experiments tested the following settings:

1. $\alpha = 0.1, \sigma = 1$
2. $\alpha = 0.1, \sigma = 4$
3. $\alpha = 0.1, \sigma = 0.25$
4. $\alpha = 0.5, \sigma = 1$
5. $\alpha = 0.9, \sigma = 1$

For each of the 5 experiments, 5 trials were done. Visualizations were made for each test, plotting a Turmite's genome's mean against the tick when it was born. I made graphs separately for each trial as well as a visualization that aggregates and color codes all the trials together, as well as a visualization that cuts the x axis off at 2.5k ticks since convergence almost always occurred by then. All of the visualizations can be found in the zip file with the code, but here are some key takeaways along with some pretty pictures of what the simulations actually looked like.

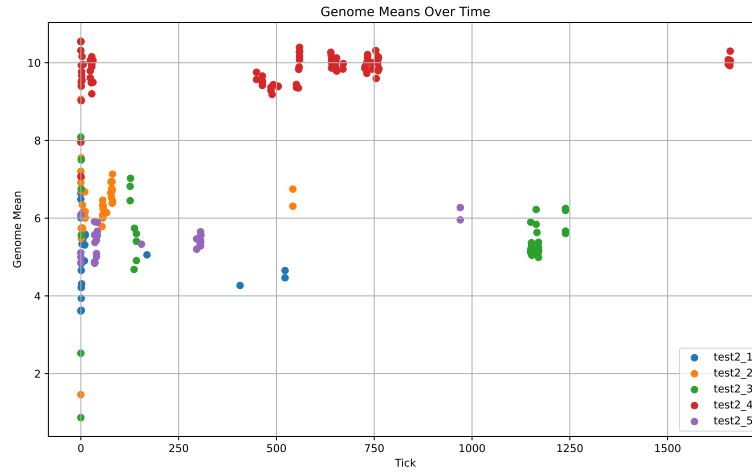


Figure 1: Test 2 ($\alpha = 0.1, \sigma = 4$) aggregated results

This visualization is unique in how it stresses the importance of the initialization of the simulation. In the red, all of the Turmites already start with their means clustering around 10, so convergence is pretty natural already. This is reinforced by the actual image from that exact simulation, notice how the colors form a coherent palette.

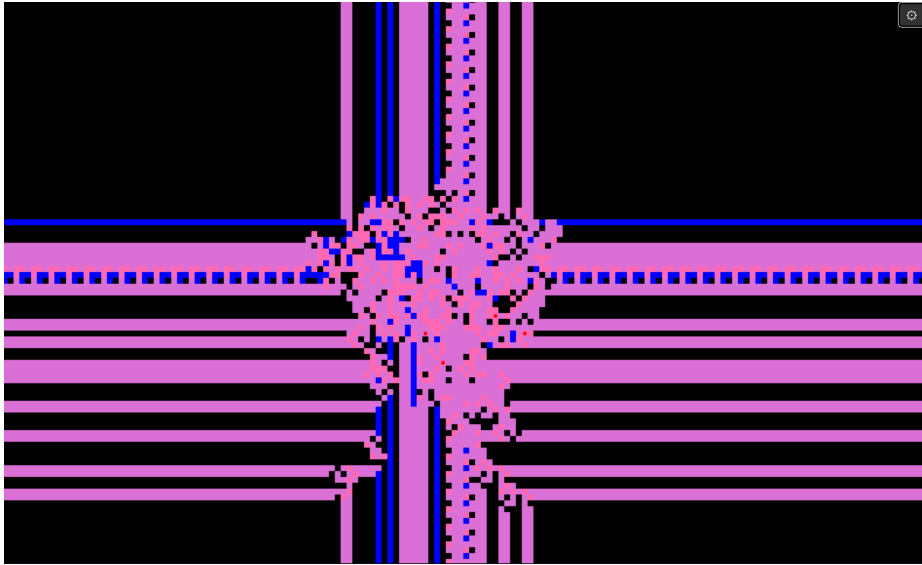


Figure 2: Test 2, Trial 4

Now, compare the heights of the clusters from the aggregated results of test 2 with those of test 3:

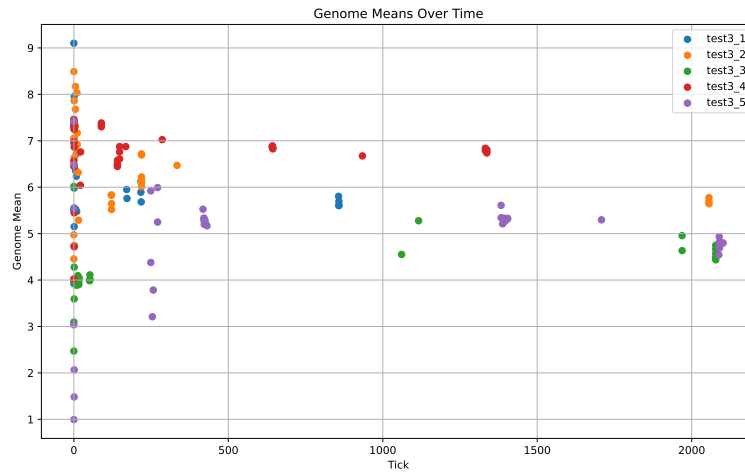


Figure 3: Test 3 ($\alpha = 0.1, \sigma = 0.25$) aggregated results

In tests 1 through 3, clusters all seem to form at various points before 500 ticks, indicating that mutation noise does not have a significant impact on convergence time. However the tightness of this convergence is clearly impacted via observations of the tightness of the clusters between tests 1 through 3.

Now observe the drastic difference in the Test 5 aggregated results:

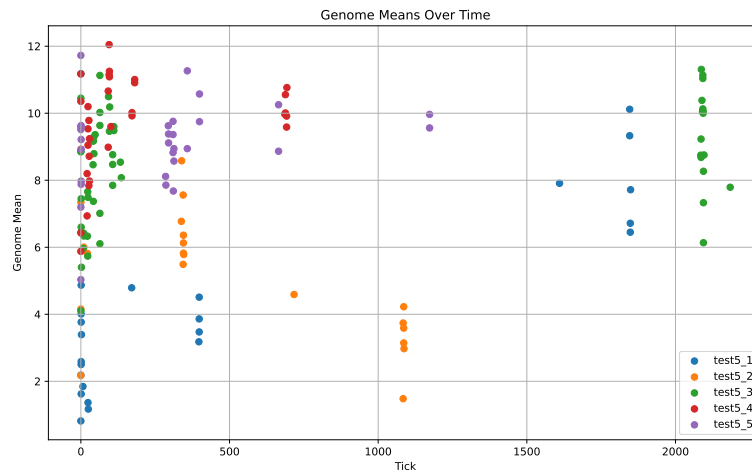
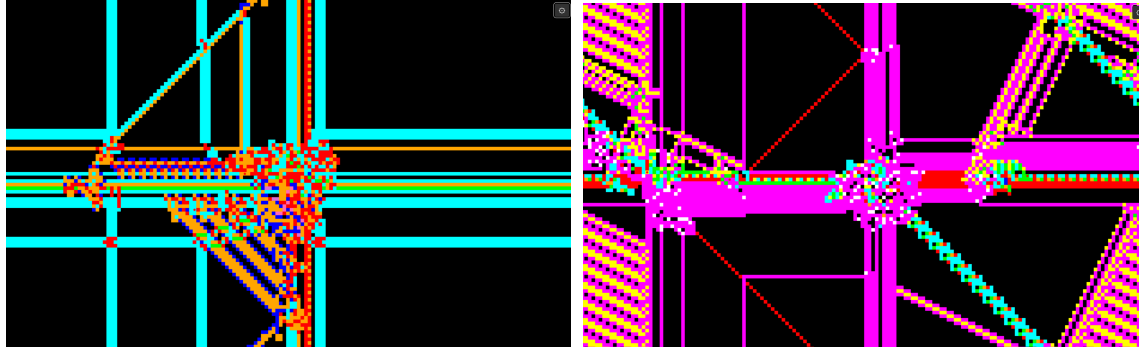


Figure 4: Test 5 ($\alpha = 0.9, \sigma = 1$) aggregated results

General clusters still manage to form, but they are so sparse and different from the initial state that it is hard to even say convergence happens. This should cause no surprises as an increase in mutation strength should cause immense amounts of noise since inheritance from one's parents is now in vicious competition with random Gaussian perturbations. Observe the following snapshots of simulations during tests 4 and 5:



(a) Test 4, Trial 2

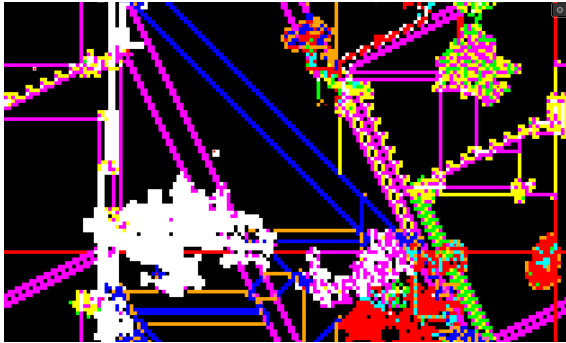
(b) Test 5, Trial 2

Clearly, these are quite a departure from the snapshot of a test 2 trial from before.

Overall the results from these experiments were rather lukewarm and for the most part met my expectations. However during these simulations I noticed that the Turmites would eventually get stuck in these infinite loops caused by all sorts of behavior. In many cases, a Turmite would run through the path of another Turmite and overwrite the colors they see, effectively boxing in the overrun Turmite. This got me thinking that speciation could be possible due to locality constraints. Turmites often would get stuck in little regions of the grid due to 1 bully Turmite that would bulldoze through the entire grid, leaving a trail of color that splits the grid into regions.

Therefore, I ran a 6th experiment with 16 Turmites instead of 4, all spawning at uniform random locations on the grid instead of the center since mating was more likely anyway. I reset the mutation parameters to my control settings, $\alpha = 0.1, \sigma = 1$.

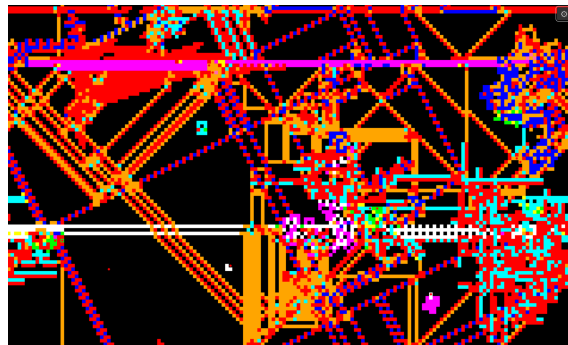
First, look at the complex ecosystems that emerge from all of these Turmites!



(a) Test 6, Trial 1



(b) Test 6, Trial 3



(c) Test 6, Trial 5

On the following page, I include the individual visualizations for each trial to illustrate the point that there does indeed seem to be speciation events happening. Note particularly the discrete clusters forming in the graphs for test6_1 and test6_5. In both graphs, we have a cluster forming at an earlier point (at roughly 1800 ticks for 6_1 and 1000 ticks for 6_5) and then a new cluster at a completely different y level later!

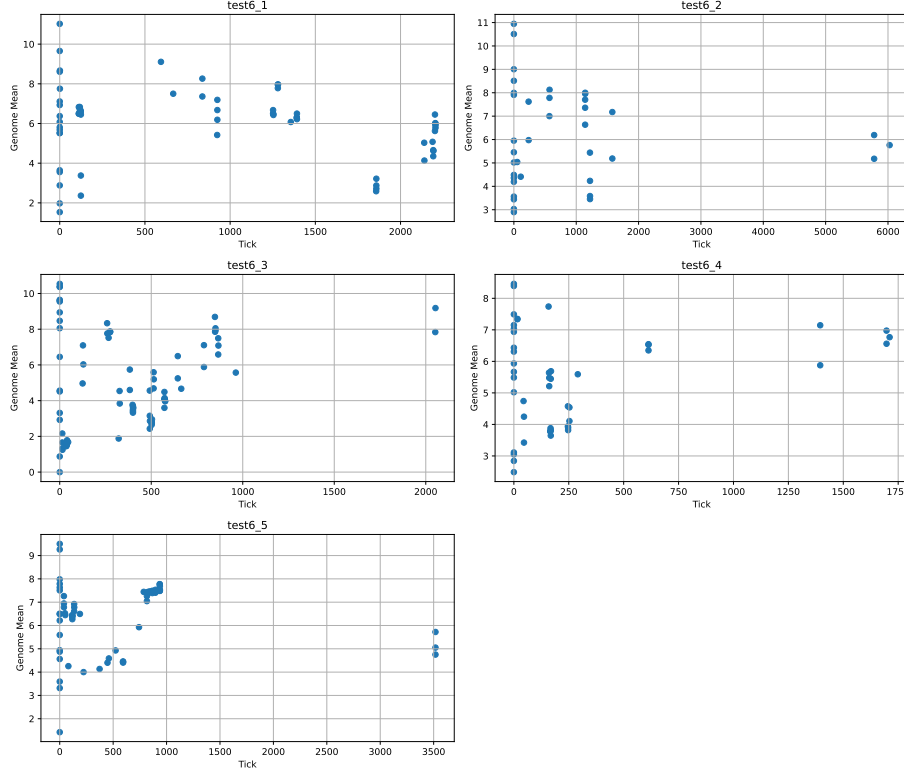


Figure 7: Test 6 ($\alpha = 0.1, \sigma = 1$) individual results

Conclusion

In this paper, I investigated whether a naive model of Turmite evolution with Gaussian genomes would converge, and if so, how the parameters of mutation strength, α , and mutation noise, σ , affect that convergence. The results largely confirmed my intuitions: since we model mutation by averaging Gaussians with Gaussian noise, we are almost guaranteed that the Turmites' genomes will converge to clusters whose widths are determined by the amount of mutation noise. Although, it is important to acknowledge that the data collected may have been biased since I chose simulations that I found interesting. Many times with 4 Turmites, they would never collide, or collide very rarely and so I discarded those trials.

The most surprising and compelling finding of these experiments was not a consequence of choosing interesting values for either of the two parameters I initially intended to alter, but rather an emergent property of scale. When the number of Turmites was increased from 4 to 16, the spatial dynamics of the grid gave rise to what appear to be genuine speciation events. Turmites partitioned the grid into

isolated regions, and within those regions, genome means converged into multiple discrete values, yielding multiple stable clusters at different points in time. This suggests that the Central Limit Theorem intuition motivating the model, that averaging should drive convergence to a single genome, breaks down due to locality constraints imposed by the behaviors of Turmites preventing truly free mixing of the variables into averages.

There are several natural directions for future work. First, it could be interesting to program behavior more closely related to how ants and termites work, with queens and nest structures and see how genetic trends play out in that scenario. Also I was considering a different mating dynamic that involved a kind of egg laying which would relax my current constraint that two Turmites must be in the exact same place at the exact same time to mate. Finally, I want to be able to adjust some of the rules by possibly adding food to the system to see if I can encourage herbivores and carnivores to exist, but the system would have been too sophisticated for this project.

References

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